

Your Mini-Capstone Project: Weeks 1–5

Pick ONE of the three projects below. Each one lets you use everything you've learned so far — Python, pandas/NumPy, SQL, exploratory data analysis, and statistics — on one real dataset and one real storyline.

Why this project matters: instead of practicing each skill in isolation, you'll use all of them together — the way a real data analyst would on the job. Choose the storyline that interests you most; all three are the same difficulty level.

How to read this guide: each project section below gives you the storyline, which skills from each week you'll practice, the exact dataset to download (with a link), a step-by-step outline to follow, and exactly what you need to submit.

Before you start — what you'll need

- A Python environment — Jupyter Notebook or Google Colab. Colab is easiest since it needs no installation and runs in your browser.
- These libraries: **pandas**, **numpy**, **matplotlib** or **seaborn**, and **scipy** (for confidence intervals). If working locally, install with: `pip install pandas numpy matplotlib seaborn scipy`. (Google Colab already has these installed.)
- A way to run SQL queries: Python's built-in `sqlite3` module needs no separate installation, or you can use a free online sandbox like DB Fiddle (dbfiddle.uk) or SQLiteOnline.com.
- A free Kaggle account (kaggle.com) to download the datasets used below.
- Confidence with what you covered in Weeks 1–5: Python basics, pandas/NumPy operations, SQL SELECT/JOIN/GROUP BY, EDA techniques, and confidence intervals — this project uses all of it, so revisit your notes for anything that feels shaky.

Project 1

E-Commerce Health Check

Difficulty: Beginner–Intermediate | Estimated time: 6–8 hours over 1–2 weeks

Storyline: You're a junior data analyst for an online retailer that wants a first health check on its operations — top products, repeat customers, order-value patterns, and whether delivery speed affects customer satisfaction.

Concepts exercised, by week

- **Week 1 (Python):** writing reusable functions for repeated calculations (e.g. a function that computes revenue per order, or flags late shipments).
- **Week 2 (Pandas/NumPy):** merging the orders, customers, and products tables; cleaning duplicates, missing values, and inconsistent formats (dates, currency).
- **Week 3 (SQL):** answering business questions directly via SQL — top products by revenue, repeat-customer rate, monthly order trends — using JOINS, GROUP BY, at least one CTE, and one window function.
- **Week 4 (EDA):** distribution of order values, correlation between shipping time and review score, outlier detection in order amounts.
- **Week 5 (Statistics):** a 95% confidence interval on average order value; an informal comparison of weekend vs. weekday order volume (sets up Week 6's formal hypothesis testing).

Dataset & prerequisites

Dataset	Brazilian E-Commerce Public Dataset by Olist (real, anonymized orders from a Brazilian marketplace, 2016-2018)
Where to get it	Kaggle: search “Olist Brazilian E-Commerce” (kaggle.com/datasets/olistbr/brazilian-ecommerce) — free download, Kaggle account required
Format	Multiple CSV files (orders, order_items, customers, products, order_reviews, payments) — ideal for practicing joins in both pandas and SQL
Size	~100,000 orders — large enough to be realistic, small enough to run on a laptop or Colab
Setup	Load the CSVs into pandas directly, and separately load them into a local SQLite database (a few lines of <code>pandas.DataFrame.to_sql()</code>) so the SQL portion can be run as real queries

Step-by-step outline

- Load all CSVs into pandas; inspect shapes, dtypes, and missing values.
- Clean: fix date formats, handle missing review scores, remove duplicate order IDs.

- Load the cleaned tables into a SQLite database.
- Write and run 4–5 SQL queries answering business questions (see above).
- Back in pandas, run EDA: plot order value distribution, check for outliers, correlate shipping time with review score.
- Compute a confidence interval for average order value; compare weekend vs. weekday order counts.
- Summarize everything into a short written report.

Deliverables

- A single Jupyter/Colab notebook containing all Python and embedded SQL code, cleanly commented.
- A short written report (1–2 pages) with 4–5 concrete findings and one business recommendation, written for a non-technical stakeholder.

Project 2

Survey Data Investigation

Difficulty: Beginner–Intermediate | Estimated time: 6–8 hours over 1–2 weeks

Storyline: You're a people-analytics researcher exploring a large survey of tech professionals, looking for patterns in compensation, experience, and job satisfaction that could inform a company's hiring or retention strategy.

Concepts exercised, by week

- **Week 1 (Python):** functions to standardize inconsistent free-text or categorical entries (e.g. normalizing job titles or country names).
- **Week 2 (Pandas/NumPy):** cleaning inconsistent categorical responses, handling missing answers, converting salary ranges or multi-select fields into usable columns.
- **Week 3 (SQL):** if the survey is split into normalized tables (respondents + responses), writing joins and aggregations across them; otherwise, load the flat file into SQLite and write GROUP BY / HAVING queries (e.g. average salary by experience bracket).
- **Week 4 (EDA):** univariate analysis of salary and satisfaction; bivariate analysis such as salary vs. years of experience, or satisfaction vs. remote-work status.
- **Week 5 (Statistics):** a confidence interval for a key metric (average salary or average satisfaction score), explained in plain English for a non-technical stakeholder.

Dataset & prerequisites

Dataset	Stack Overflow Annual Developer Survey (any recent year)
Where to get it	insights.stackoverflow.com/survey — free public download, no account required (alternative: Kaggle also mirrors past years' survey files)
Format	Single large CSV plus a schema/question reference file
Size	60,000–90,000 responses depending on year; wide (100+ columns) — good practice for selecting and cleaning only the relevant columns
Alternative dataset	Kaggle “Mental Health in Tech Survey” (kaggle.com/datasets/osmi/mental-health-in-tech-survey) — smaller and simpler, better for a lighter-weight version of this project

Step-by-step outline

- Load the survey CSV; identify and select the 10–15 columns relevant to the investigation (salary, experience, satisfaction, remote status, country, etc.).
- Clean: standardize categorical values, handle missing/skipped responses, convert salary text or ranges into numeric columns.
- Load into SQLite and write 3–4 aggregation queries (e.g. average salary by experience bracket, satisfaction by work arrangement).
- Run EDA: distributions, group comparisons, at least one correlation or cross-tab analysis.

- Compute a confidence interval for one key metric and write a plain-English interpretation.
- Summarize findings into a short report aimed at a non-technical HR or leadership audience.

Deliverables

- One notebook with cleaning, SQL, EDA, and statistics steps, clearly separated into sections.
- A short written summary (1–2 pages) translating the confidence interval and key findings into plain language, as if presenting to a non-technical stakeholder.

Project 3

City / Public Data Explainer

Difficulty: Intermediate | Estimated time: 6–9 hours over 1–2 weeks

Storyline: You're a civic-data analyst investigating a trend in open government data (crime, weather, or transit) and must determine whether an observed pattern is a real signal or just noise.

Concepts exercised, by week

- **Week 1 (Python):** a scripted ingestion pipeline — functions to load, filter by date range, and standardize the raw files.
- **Week 2 (Pandas/NumPy):** reshaping data with pivot/melt to prepare time-based or category-based tables for trend analysis.
- **Week 3 (SQL):** aggregating by time period, location, and category (e.g. incidents per month per district) using GROUP BY and window functions for running totals or period-over-period comparisons.
- **Week 4 (EDA):** visualizing the trend over time, checking seasonality, spotting anomalies or reporting gaps.
- **Week 5 (Statistics):** quantifying the trend with a mean/variance and confidence interval, and writing up whether it looks like a genuine pattern or within normal variation.

Dataset & prerequisites

Dataset options	1) Chicago Crime Data — data.cityofchicago.org (Crimes - 2001 to present) 2) NYC Taxi Trip Data (a monthly sample) — www1.nyc.gov/site/tlc/about/tlc-trip-record-data.page 3) NOAA daily weather data for any city — ncei.noaa.gov (free, no account needed for basic downloads)
Recommended pick	Chicago Crime Data is the easiest to start with: well-documented columns, downloadable as a filtered CSV (limit to 1–2 years and a few districts to keep file size manageable for a class project)
Format	CSV, typically one row per incident/trip/day
Size	Filter to a manageable slice — e.g. one city, 12–24 months — so files stay in the tens-of-thousands-of-rows range rather than millions
Setup	Load into pandas; load the same (filtered) data into SQLite for the aggregation-query portion

Step-by-step outline

- Download a filtered slice of the chosen dataset (limit rows/date range up front to keep it manageable).
- Write a small ingestion script: load, parse dates, standardize category labels.
- Load into SQLite; write aggregation queries by time period and category (e.g. monthly counts per crime type or district).

- Reshape the aggregated data with pivot/melt for easier plotting.
- Run EDA: plot the trend over time, check for seasonality or obvious anomalies.
- Quantify the trend statistically (mean, variance, confidence interval) and assess whether it's a real pattern or noise.
- Write up the conclusion in plain language, stating the confidence level of the claim.

Deliverables

- One notebook covering ingestion, SQL aggregation, reshaping, EDA, and the statistical test.
- A short written explainer (1–2 pages) stating the trend, the evidence for or against it being real, and one recommendation or open question for further investigation.

A tip for later: keep this project handy after Week 6 (Hypothesis Testing). Any pattern you noticed informally here — like the weekend-vs-weekday order difference, or the trend you investigated — is a perfect candidate to test formally once you learn hypothesis testing.

A mini-capstone covering Weeks 1–5 of your data science program. Good luck!